Laser speed gate

Final Version

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Objectives

The objective of this project was to develop a double laser tripwire system.
Using two sets of lasers and receivers, the system is able to calculate the speed with which either an object or a human interrupts set 1 and set 2 directly after each other.

Requirements

The system works in both directions, meaning both sensors are able to stop and start the timer depending on whether they were interrupted first or second. If the second set is not activated within a certain time of the first set being activated, the timer displays an error.

- The system stores the measurements and displays statistics (e.g., max/min/average of the last N runs) after the final run of the round. The user cannot toggle between the stats rather it loops infinitely around fastest, slowest and average speed and time until toggled off.
- The user can input distance, choose between units (m/s, km/h, mph) and input the amount of runs they're going to do.
- If a user passes the first sensor and turns around without passing the second sensor, the system will interrupt that as the second sensor never being passed. The system will time out and display an error.
- If a beam is blocked for too long without the second sensor also being triggered the system will time out and display an error.

Solutions

DTEK-V board with the program logic uses GPIO input from two phototransistors on a breadboard. The two lasers are on a separate breadboard with its own battery power, the lasers are pointed towards the corresponding phototransistors. The distance between the two wires is static and was placed 12 cm in between due to a limitation of the breadboards. There are diodes on the breadboard that when they go unlit signals a passage or lack of connection to the lasers. Before a lap/run there is a laser connection check that causes an error if not successful. When the phototransistor loses the light signal to either gate, that gate is considered the first

gate and the timer starts. Note that it is not when the phototransistor regains connection. The timer will end when the second wire is cut. The speed is based on the known distance and the measured time.

If the second gate is not passed in a set time an error is displayed and the lap/run have to be redone, this fulfills the last requirement as well due to how we measure as mentioned previously.

The DTEK's 7-segment display is used to display everything to the user. The program starts with a menu where the user is forced to manually choose distance, unit system and number of laps/runs. The display shows which run is being made. After all runs are completed the statistics for max/min/average are looped around for the user to see and after a long press of the button is ended and the program restarts.

Verification

Verification of the system is done manually. The time is verified by using an external stopwatch and the speed measurement through other speed measurement platforms. Since the distance and time are known, validating the speed numbers our code produces by hand is quite simple.

Contribution

Gustaf handled the external hardware setup, I/O and the I/O logic, making sure the timer starts and stops correctly. Elvis handled the speed calculation logic, 7-segment display groundwork and also made sure that the time, speed and lap status are displayed appropriately. Gustaf made bugfixes during and after hardware trials.

Reflection

The hardest part about this project was getting the hardware set up correctly, lining up the lasers to the sensors, and making sure the polling of the GPIOs were happening fast enough. The most time consuming and demanding part of the project was to make sure everything was displayed correctly on the seven segment displays. Using the second button to toggle between different parts of the program went very well. Overall a very fun project with pitfalls that were interesting to solve.